

- Number Representation and Accuracy -

Exercise 1: Binary/Floating-point representation

- (1) Give the representation of the real numbers 234 and 12.625 in the binary system.
- (2) Give the floating-point representation of the real number -3.625 in the binary system.
- (3) What is the advantage of this representation?

Exercise 2: IEEE 754 Floating-Point Standard

- (1) Represent the decimal numbers 3.14 and -8.625 using IEEE 754 Floating-Point Standard -Single Precision-
- (2) Represent the decimal number 35.5 using IEEE 754 Floating-Point Standard -Double Precision-
- (3) Give the binary representation in floating-point of the number A represented in IEEE 754 Floating-Point Standard SP:

$$A = 01000010001100110000000000000000$$

- (4) Let A and B two numbers represented in IEEE 754 Floating-Point Standard SP,
 - a. Add the numbers A and B such that:

$$A = 01000000111000000000000000000000 \quad \text{and} \quad B = 01000000000100000000000000000000$$

- b. Multiply the numbers A and B such that:

$$A = 01000000111000000000000000000000 \quad \text{and} \quad B = 01000001000110000000000000000000$$

- (5) Determine the interval of the representable real numbers using IEEE 754 Floating-Point Standard SP

Exercise 3: Significant Digits

- (1) Give the number of significant digits in the following numbers:

$$(a) 67.1 \quad (b) 0.00173 \quad (c) 2.30 \times 10^{-9} \quad (d) 3.0054 \quad (e) 0.0004000$$

(2) Rewrite the following real numbers considering the number of significant digits given between parenthesis:

- (a) 6243 (2) (b) 0.006738 (3) (c) 238.62 (3) (d) 6.345 (2)

(3) Give the results of the operations below if they are run on a computer storing the numbers in the decimal system with the following properties: $\epsilon_{\text{machine}} = 10^{-5}$, and $-20 < e < 20$ (e exponent).

- (a) $1 + 10^{-6}$ (b) $1 + 10^4$ (c) $1 + 10^6$ (d) $10^3 + 10^6$ (e) $10^6/10^{-12}$
 (f) $10^{-9} \times 10^{-16}$ (g) $10^{28} + 10^4$ (h) $10^5/10^{-21}$ (i) $\sqrt{10^{10} - 10^1}$ (j) $\ln 10^{-25}$

Exercise 4: Errors and accuracy

(1) Evaluate the **Absolute True Error** and **Absolute Percent Relative Error** committed by approximating the number π by each of the following values:

- (a) 3 (b) 3.14 (c) $\frac{22}{7}$

(2) The Area A of the earth is calculated using the following formula where r is the radius:

$$A = 4\pi r^2$$

- Give 5 approximations that are used to calculate this area on a machine, and explain the impact of each approximation on the result.
- If the radius $r = 6376$ is known with a precision of 2%, what is the absolute percent relative error on the area ?

Exercise 5: Rounding and Chopping

(1)

- Perform with two methods (using the distributivity of the multiplication on the addition) the operation $122 \times (133 + 695)$ using 3 significant digits and floating arithmetic with rounding and chopping.
- What do you observe?

(2)

- What (relative) error do we make when approximating $\cos(1.57079)$ by $\cos(1.57078)$?
- Show that the absolute true error when approximating $\cos(x)$ by $\cos(x + h)$ where h is a small perturbation is given by $e_{\text{abs}} \simeq h$ for values of x close to $\pi/2$.
- Conclude that the absolute percent relative error is $e_{\text{rel}} \simeq h \tan(x) \simeq \infty$. What do you conclude?